



Niklas Glaser

Heinrich-Heine-Universität Düsseldorf

April 2026



Overview

- 1 Motivation
- 2 Approach
- 3 Experiments
- 4 Results
- 5 Discussion
- 6 Conclusion

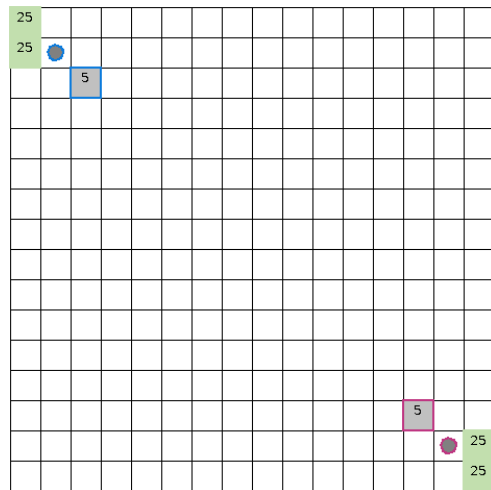
Problem Setting

Why MicroRTS?

- simplified RTS environment for RL research
- still contains resource management, unit production and combat
- supports controlled and repeatable experiments

Main challenge

- large combinatorial action space
- simultaneous decisions
- delayed and sparse rewards
- robustness against diverse opponents



Initial state of the basesWorkers16x16A map.

Motivation and Research Question

Starting point

The existing Base Agent trained with Behavioral Cloning (BC) and Proximal Policy Optimization (PPO) is already strong, but it still has weaknesses against several opponents and a limited strategic repertoire.

Goal

Improve robustness and generalization beyond a fixed set of training bots by adding Self-Play and League Training (LT).

Research question

How effective is League Training with PPO in improving the robustness and generalization of a BC/PPO-pretrained MicroRTS agent, and how strongly does this depend on the diversity of the initial agents and BC experts?

BC $\rightarrow$ BC-FT $\rightarrow$ PPO $\rightarrow$ LT $\rightarrow$ PPO fine-tuning

- **BC / BC-FT:** imitate strong scripted opponents to obtain a competent initial policy
- **PPO:** refine the policy through exploration
- **LT:** train against a population of evolving opponents instead of only bots or pure self-play
- **Fine-tuning:** fix remaining weaknesses after LT, especially against WorkerRushAI

From Self-Play to League Training

- SP** train against the current version of the agent; simple, but prone to overfitting or cyclical strategies
- FSP** train against a set of past policies; reduces forgetting, but opponent choice is still crude
- PFSP** sample opponents based on current win rates; focuses on informative and non-trivial matchups
- LT** add role-based agents: Main Agent, Historical Agents, Main Exploiters and League Exploiters

Why LT?

LT does not only preserve diversity through historical snapshots, it also creates diversity through exploiters that actively search for weaknesses of the current policy.

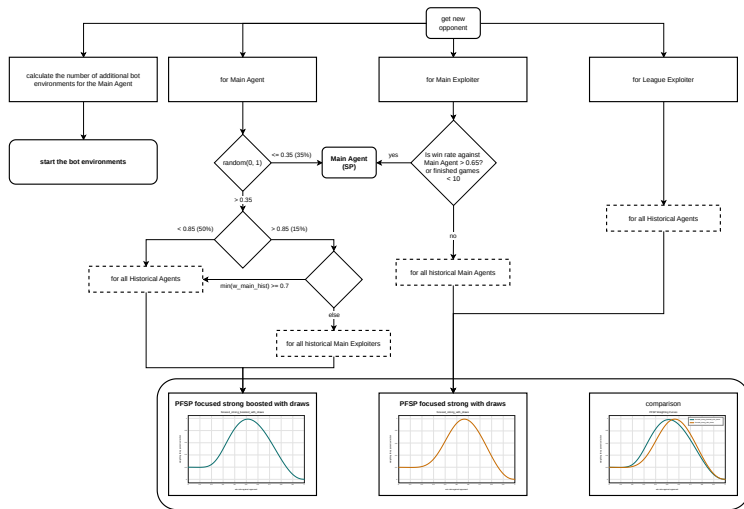
League Training Design

- **Main Agent:** primary policy
- **Historical Agents:** frozen snapshots for stability
- **Main Exploiters:** exploit weaknesses of historical Main Agents
- **League Exploiters:** search for broader league weaknesses
- **PFSP:** opponent sampling based on current win rates

Key idea

LT turns robustness against diverse opponents into part of the optimization target.

League Training Matchmaking

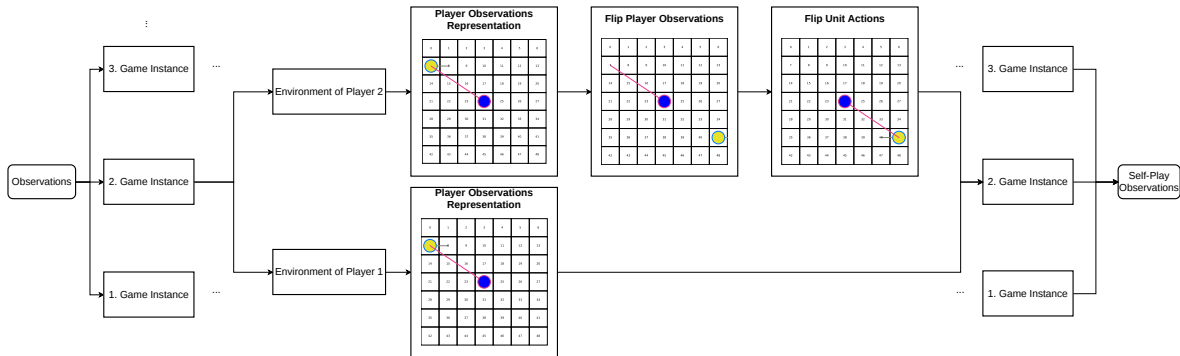


Matchmaking and opponent sampling in the League Training setup.

Important Implementation Changes

- dynamic bot environments for CoacAI and Mayari to stabilize the early LT phase
- observation adjustment for player 2 in self-play by flipping GridNet observations
- delta score reward to provide denser learning signals in MicroRTS
- annealed entropy coefficient to increase early exploration, especially for exploiters
- new initial exploiter agents with different unit-type focuses to improve diversity
-
- corrected exploiter reset and checkpoint logic compared to the AlphaStar pseudocode

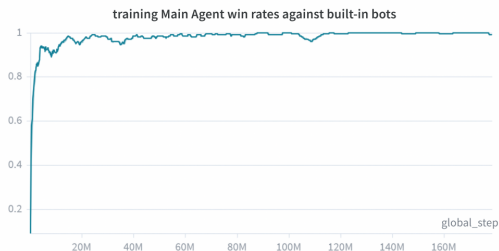
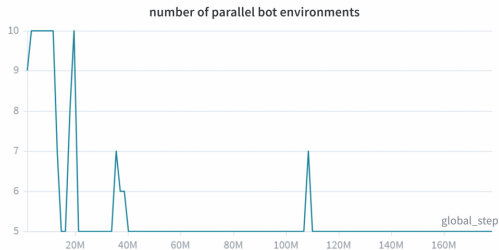
Observation Adjustment for Self-Play



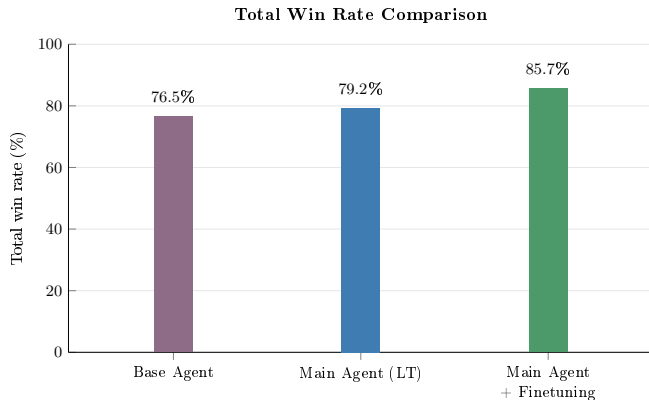
Player-2 observations are transformed so that self-play remains compatible with a Base Agent trained only as player 1.

Experiment Setup

- map: basesWorkers16x16A
- LT setup: 1 Main Agent, 4 Main Exploiters, 1 League Exploiter
- 25 environments per learning agent
- 5 to 10 additional dynamic bot environments
- approximately 130 million LT steps
- training time: about 100 hours
- maximum setup size: 160 parallel environments
- high memory cost, CPU bottleneck



Main Quantitative Result



- overall win rate improves from **76.5%** to **79.2%** after LT
- after PPO fine-tuning it reaches **85.7%**
- overall loss rate drops from **7.7%** to **2.2%**

Evaluation Results

Opponent	Main Agent without unit focus	Base Agent	Main Agent (LT)	Main Agent + Fine-tuning
<i>bots used in training</i>				
CoacAI	100.0	96.0	100.0	100.0
Mayari	100.0	67.3	100.0	100.0
<i>Built-in bots</i>				
WorkerRushAI	31.0	100.0	35.0	99.5
PassiveAI	100.0	99.7	100.0	100.0
LightRushAI	100.0	98.0	100.0	100.0
RandomAI	100.0	99.0	100.0	100.0
RandomBiasedAI	91.0	79.0	97.0	97.5
<i>Other bots</i>				
Rojo	100.0	81.0	100.0	100.0
MixedBot	45.0	43.3	57.0	60.5
Izanagi	68.0	71.7	70.0	72.5
Tiamat	94.0	93.0	99.0	99.0
Droplet	45.0	67.7	62.0	75.5
GuidedRojoA3N	64.0	74.7	87.0	88.5
NaiveMCTS	0.0	0.7	2.0	6.5
overall	74.1	76.5	79.2	85.7

Evaluation Results

Opponent	Main Agent without unit focus	Base Agent	Main Agent (LT)	Main Agent + Fine-tuning
<i>bots used in training</i>				
CoacAI	0.0	2.0	0.0	0.0
Mayari	0.0	27.3	0.0	0.0
<i>Built-in bots</i>				
WorkerRushAI	63.0	0.0	55.0	0.5
PassiveAI	0.0	0.0	0.0	0.0
LightRushAI	0.0	2.0	0.0	0.0
RandomAI	0.0	0.0	0.0	0.0
RandomBiasedAI	0.0	0.0	0.0	0.0
<i>Other bots</i>				
Rojo	0.0	5.3	0.0	0.0
MixedBot	33.0	34.3	20.0	13.5
Izanagi	15.0	16.3	7.0	6.0
Tiamat	6.0	7.0	1.0	0.5
Droplet	22.0	9.0	22.0	10.5
GuidedRojoA3N	0.0	1.7	0.0	0.0
NaiveMCTS	0.0	2.7	0.0	0.0
overall	9.9	7.7	7.5	2.2

Interpretation of the Results

- LT improves performance not only against training bots, but also against many unseen evaluation opponents.
- The Main Agent learned stronger defensive behavior while maintaining offensive pressure.
- Fine-tuning after LT largely removes the WorkerRushAI weakness and improves the overall average further.

Qualitative Findings

- LT improves more than just the average score; it changes the learned behavior.
- It performs better against draw-oriented or awkward opponents such as RandomBiasedAI.
- The unit-focused Main Agent is consistently better than the variant without unit focus.

Why Fine-Tuning Helps

Problem left after LT

WorkerRushAI remained the clearest exploitable weakness of the Main Agent after LT.

- win rate against WorkerRushAI after LT: **35.0%**
- win rate after PPO fine-tuning: **99.5%**
- the overall win rate also increases further from **79.2%** to **85.7%**

Interpretation

Fine-tuning repaired a concrete exploit without hurting generalization on the evaluated opponent set.

Player 1 Priority

Observation

MicroRTS gives player 1 action priority, so a potential concern is that self-play style training could accidentally favor player-1-specific behavior.

Result from the thesis

In one self-match evaluation of the same agent over 200 games, player 1 achieved 38% wins, 28% draws and player 2 achieved 34% wins.

- minimal effect on evaluations
- alternating the learning side to prevent the agent from exploiting player-1-specific priority

[▶ Open Player 1 Priority Example Video](#)

Base Agent vs Base Agent

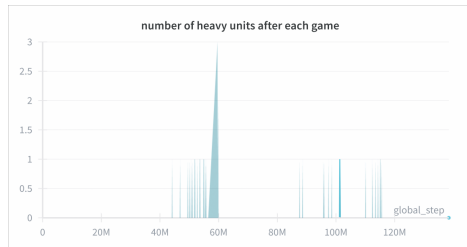
Video example for the Base Agent's preferred strategy

► Open Video

If the PDF viewer does not open the movie object, open `images/MictorTS_BaseAgent_vs_BaseAgent_edited.mp4` manually.

Main Limitation: Diversity

- Exploiters initialized from similar policies tend to discover similar weaknesses.
- Some strategically different exploits are hard to reach because they require first unlearning the Base Agent's preferred behavior.
- New initial agents with different unit focuses improved exploiter diversity and generalization.



Why New Initial Agents Matter

- exploiters initialized from similar agents tend to find similar weaknesses
- new initial agents focused on different unit types broaden the exploiter set
- this improves generalization against a wider range of opponents
- LT therefore depends strongly on the diversity already present at initialization

Scope and Remaining Limitations

- evaluation is limited to one map: `basesWorkers16x16A`
- the setting uses full observability and no fog of war
- LT requires substantially more computation than pure PPO
- exploiter diversity is still the main bottleneck

Takeaway

The method works, but the strength of the final result depends strongly on the diversity and cost of the training setup.

Conclusion

Conclusion

League Training with PPO improves the robustness and generalization of the MicroRTS Base Agent. Additional PPO fine-tuning improves the final agent even further.

Key takeaway

The effectiveness of LT depends strongly on exploiter diversity,⁵ which in turn depends on the diversity of the initial agents and BC experts.

- LT improves most matchups and the overall performance
- fine-tuning further improves the final agent
- diversity of initial agents is the critical practical factor

Future Work

- train on multiple maps and with fog of war
- improve the diversity and strength of initial agents
- reduce computational cost of MicroRTS training
- investigate stronger diversity mechanisms for exploiters

Thank you

Questions?